

1. Create user with name Techie and provide sudo access to user.

```
vboxuser@ubuntu:~/Desktop$ sudo usermod -aG sudo techie
vboxuser@ubuntu:~/Desktop$ groups techie
techie : techie sudo
vboxuser@ubuntu:~/Desktop$ +S+S+S+S=
```

2. Navigate to the home directory.

```
vboxuser@ubuntu:~/Desktop$ cd ~
vboxuser@ubuntu:~$ pwd
/home/vboxuser
vboxuser@ubuntu:~$
```

3. Create a new directory.

```
vboxuser@ubuntu:~/Desktop$ pwd
/home/vboxuser/Desktop
vboxuser@ubuntu:~/Desktop$ mkdir test
vboxuser@ubuntu:~/Desktop$ ls
test
vboxuser@ubuntu:~/Desktop$ cd test
vboxuser@ubuntu:~/Desktop/test$ pwd
/home/vboxuser/Desktop/test
vboxuser@ubuntu:~/Desktop/test$
```

4. List the contents of a directory.

```
vboxuser@ubuntu:~/Desktop$ ls -l
test
vboxuser@ubuntu:~/Desktop$ ls -la
.  ..  test
vboxuser@ubuntu:~/Desktop$
```

5. Change the current directory.

```
vboxuser@ubuntu:~/Desktop$ pwd
/home/vboxuser/Desktop
vboxuser@ubuntu:~/Desktop$ ls
test
vboxuser@ubuntu:~/Desktop$ cd test
vboxuser@ubuntu:~/Desktop/test$ ped
Command 'ped' not found, but there are 16 similar ones.
vboxuser@ubuntu:~/Desktop/test$ pwd
/home/vboxuser/Desktop/test
```

6. Create a new empty file.

```
vboxuser@ubuntu:~/Desktop/test$ touch file .txt
vboxuser@ubuntu:~/Desktop/test$ ls
file
vboxuser@ubuntu:~/Desktop/test$ touch output
vboxuser@ubuntu:~/Desktop/test$ ls
file  output
vboxuser@ubuntu:~/Desktop/test$
```

7. View the contents of a file.

```
vboxuser@ubuntu:~/Desktop/test$ pwd
/home/vboxuser/Desktop/test
vboxuser@ubuntu:~/Desktop/test$ ls
file  output
vboxuser@ubuntu:~/Desktop/test$ cat output
vboxuser@ubuntu:~/Desktop/test$
vboxuser@ubuntu:~/Desktop/test$ cat output
vboxuser@ubuntu:~/Desktop/test$ vi output
vboxuser@ubuntu:~/Desktop/test$ cat output
sample
vboxuser@ubuntu:~/Desktop/test$
```

8. Copy a file to another location.

```
vboxuser@ubuntu:~/Desktop$ ls Backup
vboxuser@ubuntu:~/Desktop$ cp -r test Backup/
vboxuser@ubuntu:~/Desktop$ ls Backup
test
vboxuser@ubuntu:~/Desktop$
```

9. Move a file to another location.

```
vboxuser@ubuntu:~/Desktop$ ls
Backup  test
vboxuser@ubuntu:~/Desktop$ ls Backup
techie  test
vboxuser@ubuntu:~/Desktop$ touch techie
vboxuser@ubuntu:~/Desktop$ ls
Backup  techie  test
vboxuser@ubuntu:~/Desktop$ mv techie Backup/
vboxuser@ubuntu:~/Desktop$ ls Backup
techie  test
vboxuser@ubuntu:~/Desktop$
```

10. Rename a file.

```
vboxuser@ubuntu:~/Desktop$ ls
Backup  test
vboxuser@ubuntu:~/Desktop$ mv Backup test
vboxuser@ubuntu:~/Desktop$ mv test techie
vboxuser@ubuntu:~/Desktop$ ls
techie
vboxuser@ubuntu:~/Desktop$
```

11. Delete a file.

```
vboxuser@ubuntu:~/Desktop$ ls techie
backup  Backup  file  output
vboxuser@ubuntu:~/Desktop$ ls -l techie
total 12
drwxrwxr-x 2 vboxuser vboxuser 4096 Aug 21 00:53 backup
drwxrwxr-x 3 vboxuser vboxuser 4096 Aug 21 03:20 Backup
-rw-rw-r-- 1 vboxuser vboxuser   0 Aug 21 00:14 file
-rw-rw-r-- 1 vboxuser vboxuser   7 Aug 21 00:35 output
vboxuser@ubuntu:~/Desktop$ mv techie/file techie/test
vboxuser@ubuntu:~/Desktop$ ls -l techie
backup
Backup
output
test
vboxuser@ubuntu:~/Desktop$
```

12. Grant or revoke permissions on a file or directory.

```
vboxuser@ubuntu:~/Desktop$ ls
techie  test
vboxuser@ubuntu:~/Desktop$ chmod u+rwx test
vboxuser@ubuntu:~/Desktop$ ls -l test
-rwxrwxr-x 1 vboxuser vboxuser 0 Aug 21 04:16 test
vboxuser@ubuntu:~/Desktop$ chmod u-w test
vboxuser@ubuntu:~/Desktop$ ls -l test
-r-xrwxr-x 1 vboxuser vboxuser 0 Aug 21 04:16 test
vboxuser@ubuntu:~/Desktop$
```

13. View the current date and time.

```
vboxuser@ubuntu:~/Desktop$ date
Fri Aug 21 04:52:10 AM PDT 2026
vboxuser@ubuntu:~/Desktop$
```

14. Check the system uptime.

```
vboxuser@ubuntu:~/Desktop$ uptime
05:02:32 up 2:51, 1 user, load average: 0.20, 0.08, 0.03
vboxuser@ubuntu:~/Desktop$
```

15. View the running processes

```
vboxuser@ubuntu:~/Desktop$ ps
  PID TTY          TIME CMD
 3355 pts/1        00:00:00 bash
 3368 pts/1        00:00:00 ps
vboxuser@ubuntu:~/Desktop$
```

16. Kill a running process.

```
vboxuser@ubuntu:~/Desktop$ ps
  PID TTY          TIME CMD
 2360 pts/0        00:00:00 bash
 2798 pts/0        00:00:00 sleep
 2843 pts/0        00:00:00 ps
vboxuser@ubuntu:~/Desktop$ kill 2798
vboxuser@ubuntu:~/Desktop$ ps
  PID TTY          TIME CMD
 2360 pts/0        00:00:00 bash
 2844 pts/0        00:00:00 ps
[1]+  Terminated                  sleep 1000
vboxuser@ubuntu:~/Desktop$
```

17. Install a package using the package manager (e.g., apt or yum).

```

Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
449 packages can be upgraded. Run 'apt list --upgradable' to see them.
vboxuser@ubuntu:~/Desktop$ sudo apt install curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcurl4
The following NEW packages will be installed:
  curl
The following packages will be upgraded:
  libcurl4
1 upgraded, 1 newly installed, 0 to remove and 448 not upgraded.
Need to get 486 kB of archives.
After this operation, 458 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcurl4 amd64 7.81.0-1ubuntu1.26 [292 kB]
Get:2 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 curl amd64 7.81.0-1ubuntu1.26 [194 kB]
Fetched 486 kB in 2s (262 kB/s)
(Reading database ... 204651 files and directories currently installed.)
Preparing to unpack .../libcurl4_7.81.0-1ubuntu1.26_amd64.deb ...
Unpacking libcurl4:amd64 (7.81.0-1ubuntu1.26) over (7.81.0-1ubuntu1.17) ...
Selecting previously unselected package curl.
Preparing to unpack .../curl_7.81.0-1ubuntu1.26_amd64.deb ...
Unpacking curl (7.81.0-1ubuntu1.26) ...
Setting up libcurl4:amd64 (7.81.0-1ubuntu1.26) ...
Setting up curl (7.81.0-1ubuntu1.26) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
vboxuser@ubuntu:~/Desktop$ curl --version

```

18. Update the system packages.

```

vboxuser@ubuntu:~/Desktop$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
vboxuser@ubuntu:~/Desktop$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
vboxuser@ubuntu:~/Desktop$ S

```

19. Create a symbolic link.

```
vboxuser@ubuntu:~/Desktop$ ll
total 16
drwxr-xr-x  4 vboxuser vboxuser 4096 Aug 22 05:19 ./
drwxr-x--- 18 vboxuser vboxuser 4096 Aug 20 05:30 ../
-rw-rw-r--  1 vboxuser vboxuser   0 Aug 22 05:07 parvez
lrwxrwxrwx  1 vboxuser vboxuser   6 Aug 22 05:08 parvez_link -> parvez
drwxrwxr-x  2 vboxuser vboxuser 4096 Aug 22 05:17 softlink/
lrwxrwxrwx  1 vboxuser vboxuser   9 Aug 22 05:19 softlinkcreated -> softlink//
drwxrwxr-x  4 vboxuser vboxuser 4096 Aug 22 05:07 techie/
-r-xrwxr-x  1 vboxuser vboxuser   0 Aug 21 04:16 test*
-rw-rw-r--  1 vboxuser vboxuser   0 Aug 21 10:57 testfile.txt
vboxuser@ubuntu:~/Desktop$ S
```

20. Search for files using the find command.

```
vboxuser@ubuntu:~/Desktop$ pwd
/home/vboxuser/Desktop
vboxuser@ubuntu:~/Desktop$ cd ~/Desktop
vboxuser@ubuntu:~/Desktop$ ls
parvez parvez_link softlink softlinkcreated techie test testfile.txt
vboxuser@ubuntu:~/Desktop$ find . -name "test"
./techie/Backup/test
./techie/test
./test
vboxuser@ubuntu:~/Desktop$ ./test
vboxuser@ubuntu:~/Desktop$
```

21. Compress and decompress files using tar.

```
vboxuser@ubuntu:~/Desktop$ ls
parvez parvez_link softlink softlinkcreated techie test testfile.txt
vboxuser@ubuntu:~/Desktop$ tar -cvf testfile.tar testfile.txt
testfile.txt
vboxuser@ubuntu:~/Desktop$ ls
parvez parvez_link softlink softlinkcreated techie test testfile.tar testfile.txt
vboxuser@ubuntu:~/Desktop$ tar -xvf testfile.tar
testfile.txt
vboxuser@ubuntu:~/Desktop$ ls
parvez parvez_link softlink softlinkcreated techie test testfile.tar testfile.txt
vboxuser@ubuntu:~/Desktop$
```

22. Monitor system resources with top or htop.

```

top - 05:42:05 up 42 min, 1 user, load average: 0.05, 0.07, 0.03
Tasks: 201 total, 1 running, 200 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.0 sy, 0.0 ni, 99.6 id, 0.2 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 1967.8 total, 82.8 free, 870.9 used, 1014.1 buff/cache
MiB Swap: 2680.0 total, 2678.7 free, 1.3 used. 904.6 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1430	vboxuser	20	0	4659368	347340	134560	S	2.0	17.2	0:27.01	gnome-shell
438	systemd+	20	0	14836	6876	6044	S	0.3	0.3	0:01.38	systemd-oomd
2506	vboxuser	20	0	555592	54504	41912	S	0.3	2.7	0:02.22	gnome-terminal-
1	root	20	0	166664	11988	8360	S	0.0	0.6	0:01.25	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.02	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_g
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_p
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
12	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_pe
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
16	root	20	0	0	0	0	S	0.0	0.0	0:00.07	ksoftirqd/0
17	root	20	0	0	0	0	I	0.0	0.0	0:00.26	rcu_preempt
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.02	migration/0
19	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
23	root	rt	0	0	0	0	S	0.0	0.0	0:00.19	migration/1
24	root	20	0	0	0	0	S	0.0	0.0	0:00.08	ksoftirqd/1
26	root	0	-20	0	0	0	I	0.0	0.0	0:00.11	kworker/1:0H-kblockd
27	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/2

23. Create and manage user groups.

```

vboxuser@ubuntu:~/Desktop$ sudo groupadd devops
groupadd: group 'devops' already exists
vboxuser@ubuntu:~/Desktop$ getent group devops
devops:x:1003:techie
vboxuser@ubuntu:~/Desktop$ sudo usermod -aG devops techie
vboxuser@ubuntu:~/Desktop$ groups techie
techie : techie sudo devops
vboxuser@ubuntu:~/Desktop$ sudo gpasswd -d techie devops
Removing user techie from group devops
vboxuser@ubuntu:~/Desktop$ groups techie
techie : techie sudo
vboxuser@ubuntu:~/Desktop$

```

24. Set up SSH password less authentication.

```
vboxuser@ubuntu: ~/Desktop$ ssh vboxuser@localhost
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-138-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

vboxuser@ubuntu:~$ ssh vboxuser@localhost
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-138-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sun Aug 23 06:59:36 2026 from 127.0.0.1
vboxuser@ubuntu:~$
```

25. Monitor log files using tail or Grep.

```

org.freedesktop.NetworkManager[563]: <Info> [1787497964.7483] failed to open /run/network/lfstate
Aug 23 08:12:44 ubuntu systemd[1]: Started Download data for packages that failed at package install time.
Aug 23 08:12:45 ubuntu gnome-session[841]: gnome-session-binary[841]: Glib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
Aug 23 08:12:45 ubuntu gnome-session[841]: gnome-session-binary[841]: Glib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
Aug 23 08:12:45 ubuntu gnome-session-binary[841]: Glib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
Aug 23 08:12:45 ubuntu gnome-session-binary[841]: Glib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
Aug 23 08:12:46 ubuntu org.gnome.Shell.desktop[878]: libEGL warning: egl: failed to create dri2 screen
Aug 23 08:12:46 ubuntu org.gnome.Shell.desktop[878]: libEGL warning: egl: failed to create dri2 screen
Aug 23 08:12:46 ubuntu org.gnome.Shell.desktop[920]: libEGL warning: egl: failed to create dri2 screen
Aug 23 08:12:46 ubuntu org.gnome.Shell.desktop[920]: libEGL warning: egl: failed to create dri2 screen
Aug 23 08:12:48 ubuntu /usr/libexec/gdm-wayland-session[840]: dbus-daemon[840]: [session uid=128 pid=840] Activated service 'org.freedesktop.systemd1' failed: Process org.freedesktop.systemd1 exited with status 1
Aug 23 08:12:49 ubuntu colord[1111]: failed to get edid data: EDID length is too small
Aug 23 08:12:49 ubuntu gsd-color[950]: failed to get edid: unable to get EDID for output
Aug 23 08:13:02 ubuntu gnome-shell[1451]: libEGL warning: egl: failed to create dri2 screen
Aug 23 08:13:02 ubuntu gnome-shell[1451]: libEGL warning: egl: failed to create dri2 screen
Aug 23 08:13:04 ubuntu systemd[1292]: Dependency failed for GNOME XSettings service.
Aug 23 08:13:04 ubuntu systemd[1292]: org.gnome.SettingsDaemon.XSettings.service: Job org.gnome.SettingsDaemon.XSettings.service/start failed with result 'dependency'.
Aug 23 08:13:04 ubuntu systemd[1292]: gnome-session-x11-services-ready.target: Job gnome-session-x11-services-ready.target/verify-active failed with result 'dependency'.
Aug 23 08:13:07 ubuntu gsd-color[1625]: failed to get edid: unable to get EDID for output
Aug 23 08:13:08 ubuntu gdm-launch-environment]: Glib-GObject: g_object_unref: assertion 'G_IS_OBJECT (object)' failed
Aug 23 08:13:10 ubuntu pulseaudio[1301]: GetManagedObjects() failed: org.freedesktop.DBus.Error.TimedOut: Failed to activate service 'org.bluez': timed out (service_start_timeout=25000ms)
Aug 23 08:13:13 ubuntu snap-store[1783]: fwupd_client_download_bytes_async: assertion 'url != NULL' failed
Aug 23 08:13:13 ubuntu snap-store[1783]: fwupd_client_download_bytes_async: assertion 'url != NULL' failed
Aug 23 08:14:14 ubuntu snap-store[1783]: fwupd_client_download_bytes_async: assertion 'url != NULL' failed
Aug 23 08:17:44 ubuntu systemd[1]: Starting Download data for packages that failed at package install time...
Aug 23 08:17:44 ubuntu systemd[1]: Finished Download data for packages that failed at package install time.
Aug 23 09:07:51 ubuntu snap-store[1783]: fwupd_client_download_bytes_async: assertion 'url != NULL' failed
Aug 23 09:07:54 ubuntu snap-store[1783]: message repeated 3 times: [ fwupd_client_download_bytes_async: assertion 'url != NULL' failed]
vboxuser@ubuntu:~/var/log$

```

26. Set up a web server (e.g., Apache or Nginx).

```

vboxuser@ubuntu:~/Desktop$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-08-23 09:53:53 PDT; 21min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3308 (apache2)
    Tasks: 55 (limit: 2259)
   Memory: 3.8M
      CPU: 105ms
   CGroup: /system.slice/apache2.service
           └─3308 /usr/sbin/apache2 -k start
             └─3309 /usr/sbin/apache2 -k start
               └─3310 /usr/sbin/apache2 -k start

Aug 23 09:53:53 ubuntu systemd[1]: Starting The Apache HTTP Server...
Aug 23 09:53:53 ubuntu systemd[1]: Started The Apache HTTP Server.
vboxuser@ubuntu:~/Desktop$ q
Command 'q' not found, but can be installed with:
sudo snap install q # version 1.6.3-1, or
sudo apt install python3-q-text-as-data # version 3.1.6-1
See 'snap info q' for additional versions.
vboxuser@ubuntu:~/Desktop$ Q
Q: command not found
vboxuser@ubuntu:~/Desktop$ curl http://localhost
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<!--
  Modified from the Debian original for Ubuntu
  Last updated: 2022-03-22
  See: https://launchpad.net/bugs/1966004
-->

```

Last updated: 2022-03-22
See: <https://launchpad.net/bugs/1966004>

```
-->
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
  <title>Apache2 Ubuntu Default Page: It works</title>
  <style type="text/css" media="screen">
* {
  margin: 0px 0px 0px 0px;
  padding: 0px 0px 0px 0px;
}

body, html {
  padding: 3px 3px 3px 3px;

  background-color: #D8DBE2;

  font-family: Ubuntu, Verdana, sans-serif;
  font-size: 11pt;
  text-align: center;
}

div.main_page {
  position: relative;
  display: table;

  width: 800px;

  margin-bottom: 3px;
  margin-left: auto;
  margin-right: auto;
```

```

located elsewhere (such as in <tt>/srv</tt>) you may need to whitelist your
document root directory in <tt>/etc/apache2/apache2.conf</tt>.
</p>
<p>
The default Ubuntu document root is <tt>/var/www/html</tt>. You
can make your own virtual hosts under /var/www.
</p>
</div>

<div class="section_header">
  <div id="bugs"></div>
    Reporting Problems
  </div>
  <div class="content_section_text">
    <p>
      Please use the <tt>ubuntu-bug</tt> tool to report bugs in the
      Apache2 package with Ubuntu. However, check <a
      href="https://bugs.launchpad.net/ubuntu/+source/apache2"
      rel="nofollow">existing bug reports</a> before reporting a new bug.
    </p>
    <p>
      Please report bugs specific to modules (such as PHP and others)
      to their respective packages, not to the web server itself.
    </p>
  </div>
</div>
</div>
<div class="validator">
</div>
</body>
</html>
vboxuser@ubuntu:~/Desktop$ sudo systemctl is-active apache2
active
vboxuser@ubuntu:~/Desktop$

```

27. Configure and secure a MySQL Database.

Success.

Reloading the privilege tables will ensure that all changes made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : y
Success.

All done!

vboxuser@ubuntu:~/Desktop\$ mysql --version

mysql Ver 8.0.46-0ubuntu0.22.04.3 for Linux on x86_64 ((Ubuntu))

vboxuser@ubuntu:~/Desktop\$ sudo mysql

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 10

Server version: 8.0.46-0ubuntu0.22.04.3 (Ubuntu)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;

```
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
```

4 rows in set (0.04 sec)

mysql> █

28. Set up a Application Server (e.g.,Apache Tomcat).

```

vboxuser@ubuntu:~/Desktop$ ls -l /var/lib/tomcat9/webapps
total 4
drwxr-xr-x 3 root root 4096 Aug 24 01:24 ROOT
vboxuser@ubuntu:~/Desktop$ ls -l /var/lib/tomcat9/webapps/ROOT
total 8
-rw-r--r-- 1 root root 1895 Aug 24 01:24 index.html
drwxr-xr-x 2 root root 4096 Aug 24 01:24 META-INF
vboxuser@ubuntu:~/Desktop$ sudo systemctl status tomcat9
[sudo] password for vboxuser:
● tomcat9.service - Apache Tomcat 9 Web Application Server
   Loaded: loaded (/lib/systemd/system/tomcat9.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-08-24 04:22:33 PDT; 6min ago
     Docs: https://tomcat.apache.org/tomcat-9.0-doc/index.html
   Process: 677 ExecStartPre=/usr/libexec/tomcat9/tomcat-update-policy.sh (code=exited, status=0/SUCCESS)
   Main PID: 712 (java)
     Tasks: 35 (limit: 2259)
    Memory: 79.4M
       CPU: 11.566s
    CGroup: /system.slice/tomcat9.service
            └─712 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.file=/var/lib/tomcat9/conf/logging.properties -Djava.util.

Aug 24 04:22:37 ubuntu tomcat9[712]: OpenSSL successfully initialized [OpenSSL 3.0.2 15 Mar 2022]
Aug 24 04:22:38 ubuntu tomcat9[712]: Initializing ProtocolHandler [http-nio-8080]
Aug 24 04:22:38 ubuntu tomcat9[712]: Server initialization in [2802] milliseconds
Aug 24 04:22:38 ubuntu tomcat9[712]: Starting service [Catalina]
Aug 24 04:22:38 ubuntu tomcat9[712]: Starting Servlet engine: [Apache Tomcat/9.0.58 (Ubuntu)]
Aug 24 04:22:38 ubuntu tomcat9[712]: Deploying web application directory [/var/lib/tomcat9/webapps/ROOT]
Aug 24 04:22:42 ubuntu tomcat9[712]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for
Aug 24 04:22:42 ubuntu tomcat9[712]: Deployment of web application directory [/var/lib/tomcat9/webapps/ROOT] has finished in [3,388] ms
Aug 24 04:22:42 ubuntu tomcat9[712]: Starting ProtocolHandler [http-nio-8080]
Aug 24 04:22:42 ubuntu tomcat9[712]: Server startup in [3685] milliseconds
vboxuser@ubuntu:~/Desktop$ curl http://localhost:8080
<?xml version="1.0" encoding="ISO-8859-1"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en">

```

29.create a service file for Apache Tomcat.(Should execute by using systemctl command).

```

apache-tomcat-10.1.59/bin/ciphers.sh
apache-tomcat-10.1.59/bin/configtest.sh
apache-tomcat-10.1.59/bin/daemon.sh
apache-tomcat-10.1.59/bin/digest.sh
apache-tomcat-10.1.59/bin/makebase.sh
apache-tomcat-10.1.59/bin/migrate.sh
apache-tomcat-10.1.59/bin/setclasspath.sh
apache-tomcat-10.1.59/bin/shutdown.sh
apache-tomcat-10.1.59/bin/startup.sh
apache-tomcat-10.1.59/bin/tool-wrapper.sh
apache-tomcat-10.1.59/bin/version.sh
[ec2-user@ip-172-31-93-125 opt]$ sudo mv apache-tomcat-10.1.59 tomcat
[ec2-user@ip-172-31-93-125 opt]$ ls -l /opt/tomcat
total 180
-rw-r-----. 1 root root 26826 Aug 13 16:59 BUILDING.txt
-rw-r-----. 1 root root 8626 Aug 13 16:59 CONTRIBUTING.md
-rw-r-----. 1 root root 60393 Aug 13 16:59 LICENSE
-rw-r-----. 1 root root 2333 Aug 13 16:59 NOTICE
-rw-r-----. 1 root root 3231 Aug 13 16:59 README.md
-rw-r-----. 1 root root 6776 Aug 13 16:59 RELEASE-NOTES
-rw-r-----. 1 root root 16114 Aug 13 16:59 RUNNING.txt
drwxr-x---. 2 root root 16384 Aug 26 06:52 bin
drwxr-x---. 2 root root 16384 Aug 13 16:59 conf
drwxr-x---. 2 root root 16384 Aug 26 06:52 lib
drwxr-x---. 2 root root 6 Aug 13 16:59 logs
drwxr-x---. 2 root root 30 Aug 26 06:52 temp
drwxr-x---. 7 root root 81 Aug 13 16:59 webapps
drwxr-x---. 2 root root 6 Aug 13 16:59 work
[ec2-user@ip-172-31-93-125 opt]$ sudo systemctl start tomcat
Failed to start tomcat.service: Unit tomcat.service not found.
[ec2-user@ip-172-31-93-125 opt]$ sudo nano /etc/systemd/system/tomcat.service
[ec2-user@ip-172-31-93-125 opt]$ sudo systemctl daemon-reload
[ec2-user@ip-172-31-93-125 opt]$ sudo systemctl start tomcat
[ec2-user@ip-172-31-93-125 opt]$ sudo systemctl status tomcat
-bash: sudu: command not found
[ec2-user@ip-172-31-93-125 opt]$ sudo systemctl start tomcat
[ec2-user@ip-172-31-93-125 opt]$ sudo systemctl status tomcat
● tomcat.service - Apache Tomcat Server
   Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-08-26 07:18:33 UTC; 2min 1s ago
     Process: 28454 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
    Main PID: 28469 (java)
      Tasks: 29 (limit: 1113)
     Memory: 77.2M
        CPU: 2.950s
    CGroup: /system.slice/tomcat.service

```

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Apache Tomcat/10.1.59



If you're seeing this, you've successfully installed Tomcat. Congratulations!

Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

[Server Status](#)
[Manager App](#)
[Host Manager](#)

Developer Quick Start

[Tomcat Setup](#)
[Realms & AAA](#)
[Examples](#)
[Servlet Specifications](#)

[First Web Application](#)
[JDBC DataSources](#)
[Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 10.1 access to the manager application is split between different users. [Read more...](#)

[Release Notes](#)

[Changelog](#)

Documentation

[Tomcat 10.1 Documentation](#)

[Tomcat 10.1 Configuration](#)

[Tomcat Wiki](#)

Find additional important configuration information in:

```
$CATALINA_HOME/RUNNING.txt
```

Developers may be interested in:

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

[tomcat-announce](#)
 Important announcements, releases, security vulnerability notifications. (Low volume).

[tomcat-users](#)
 User support and discussion

[taglibs-user](#)
 User support and discussion for [Apache Taglibs](#)

30. Print specific columns from a delimited file.

```
vboxuser@ubuntu:~/Desktop$ cd~/desktop
bash: cd~/desktop: No such file or directory
vboxuser@ubuntu:~/Desktop$ pwd
/home/vboxuser/Desktop
vboxuser@ubuntu:~/Desktop$ ls
parvez  parvez_link  softlink  softlinkcreated  techie  test  testfile.tar  testfile.txt
vboxuser@ubuntu:~/Desktop$ mkdir spider
vboxuser@ubuntu:~/Desktop$ ls
parvez  parvez_link  softlink  softlinkcreated  spider  techie  test  testfile.tar  testfile.txt
vboxuser@ubuntu:~/Desktop$ cd spider
vboxuser@ubuntu:~/Desktop/spider$ pwd
/home/vboxuser/Desktop/spider
vboxuser@ubuntu:~/Desktop/spider$ nano students.txt
vboxuser@ubuntu:~/Desktop/spider$ cat students.txt
101,Parvez,finance
102,Ahmad,IT
103,Rahul,HR
104,John,Marketing==
vboxuser@ubuntu:~/Desktop/spider$ cut -d',' -f2 students.txt
Parvez
Ahmad
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$ cut -d',' -f3 students.txt
finance
IT
HR
Marketing==
vboxuser@ubuntu:~/Desktop/spider$
```

31. Filter and print lines based on a specific pattern or condition.

```
File Machine View Input Devices Help
boxuser@ubuntu:~/Desktop$ pwd
home/vboxuser/Desktop
boxuser@ubuntu:~/Desktop$ cd spider
boxuser@ubuntu:~/Desktop/spider$ ls
tudents.txt
boxuser@ubuntu:~/Desktop/spider$ cat student.txt
at: student.txt: No such file or directory
boxuser@ubuntu:~/Desktop/spider$ cat students.txt
01,Parvez,finance
02,Ahmad,IT
03,Rahul,HR
04,John,Marketing==
boxuser@ubuntu:~/Desktop/spider$ grep "parvez" students.txt
boxuser@ubuntu:~/Desktop/spider$ ls
tudents.txt
boxuser@ubuntu:~/Desktop/spider$ cat students.txt
01,Parvez,finance
02,Ahmad,IT
03,Rahul,HR
04,John,Marketing==
boxuser@ubuntu:~/Desktop/spider$ grep "Parvez" students.txt
01,Parvez,finance
boxuser@ubuntu:~/Desktop/spider$ grep "IT" students.txt
02,Ahmad,IT
boxuser@ubuntu:~/Desktop/spider$
```

32. Calculate and print the average, sum, or other statistics of a column.

```
vboxuser@ubuntu:~/Desktop/spider$ awk -F',' '{sum += $1} END {print sum}' students.txt
410
vboxuser@ubuntu:~/Desktop/spider$ awk -F',' '{sum += $1} END {print sum/NR}' students.txt
102.5
vboxuser@ubuntu:~/Desktop/spider$
```

33. Perform string manipulation, such as extracting substrings or changing case.

```
vboxuser@ubuntu:~/Desktop/spider$ cd /home/vboxuser/Desktop/spider
vboxuser@ubuntu:~/Desktop/spider$ nano vinom.txt
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
Parvez
Sahil
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$ awk '{print substr($0,1,3)}' vinom.txt
Par
Sah
Rah
Joh
vboxuser@ubuntu:~/Desktop/spider$ awk '{print toupper($0)}' vinom.txt
PARVEZ
SAHIL
RAHUL
JOHN
vboxuser@ubuntu:~/Desktop/spider$ awk '{print lower($0)}' vinom.txt
awk: line 2: function lower never defined
vboxuser@ubuntu:~/Desktop/spider$ awk '{print tolower($0)}' vinom.txt
parvez
sahil
rahul
john
vboxuser@ubuntu:~/Desktop/spider$
```

34. Count the occurrences of a specific pattern in a file.

```
vboxuser@ubuntu:~/Desktop/spider$ cd /home/vboxuser/Desktop/spider
vboxuser@ubuntu:~/Desktop/spider$ ls
students.txt  vinom.txt
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
Parvez
Sahil
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$ grep -c "Parvez" vinom.txt
1
vboxuser@ubuntu:~/Desktop/spider$ grep -c "john" vinom.txt
0
vboxuser@ubuntu:~/Desktop/spider$ grep -c "Jojn" vinom.txt
0
vboxuser@ubuntu:~/Desktop/spider$ grep -c "Sahli" vinom.txt
0
vboxuser@ubuntu:~/Desktop/spider$ █
```

35. Sort lines based on a specific field or column.

```
vboxuser@ubuntu:~/Desktop/spider$ cd /home/vboxuser/Desktop/spider
vboxuser@ubuntu:~/Desktop/spider$ ls
students.txt  vinom.txt
vboxuser@ubuntu:~/Desktop/spider$ students.txt
students.txt: command not found
vboxuser@ubuntu:~/Desktop/spider$ Students.txt
Students.txt: command not found
vboxuser@ubuntu:~/Desktop/spider$ cat students.txt
101,Parvez,finance
102,Ahmad,IT
103,Rahul,HR
104,John,Marketing==
vboxuser@ubuntu:~/Desktop/spider$ sort -t',' -k2 students.txt
102,Ahmad,IT
104,John,Marketing==
101,Parvez,finance
103,Rahul,HR
vboxuser@ubuntu:~/Desktop/spider$
```

36. Merge multiple files based on a common field or column.

```
vboxuser@ubuntu:~/Desktop/spider$ cd /home/vboxuser/Desktop/spider
vboxuser@ubuntu:~/Desktop/spider$ pwd
/home/vboxuser/Desktop/spider
vboxuser@ubuntu:~/Desktop/spider$ nano studentsl.txt
vboxuser@ubuntu:~/Desktop/spider$ nano marks.txt
vboxuser@ubuntu:~/Desktop/spider$ ls
marks.txt  studentsl.txt  students_sorted.txt  students.txt  vinom.txt
vboxuser@ubuntu:~/Desktop/spider$ sort -t',' -k1,l studentsl.txt >students_sorted.txt
sort: invalid number at field start: invalid count at start of 'l,l'
vboxuser@ubuntu:~/Desktop/spider$ sort -t',' -k1,1 studentsl.txt >students_sorted.txt
vboxuser@ubuntu:~/Desktop/spider$ sort -t',' -k1,1 marks.txt >marks_sorted.txt
vboxuser@ubuntu:~/Desktop/spider$ join -t',' -1 1 -2 1 students_sorted.txt marks_sorted.txt
101,Parvez,80
102,Ahmad,70
103,Rahul,90
104,John,60
vboxuser@ubuntu:~/Desktop/spider$
```

37. Substitute text in a file using search and replace.

```
vboxuser@ubuntu:~/Desktop/spider$ cd /home/vboxuser/Desktop/spider
vboxuser@ubuntu:~/Desktop/spider$ ls
marks_sorted.txt  marks.txt  studentsl.txt  students_sorted.txt  students.txt  vinom.txt
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
Parvez
Sahil
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$ sed -i 's/Parvez/Mohd/g' vinom.txt
sed: -e expression #1, char 16: unknown option to `s'
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
Parvez
Sahil
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$ sed 's/Parvez/Mohd/g' vinom.txt
Mohd
Sahil
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$
```

38. Delete specific lines based on a pattern or line number.

```
vboxuser@ubuntu:~/Desktop/spider$ cat /home/vboxuser/Desktop/spider
cat: /home/vboxuser/Desktop/spider: Is a directory
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
Parvez
Sahil
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$ sed -i '2d' vinom.txt
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
Parvez
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$
```

39. Append or insert text before or after a specific pattern or line.

```

vboxuser@ubuntu:~/Desktop/spider$ cd /home/vboxuser/Desktop/spider
vboxuser@ubuntu:~/Desktop/spider$ ls
marks_sorted.txt  marks.txt  studentsl.txt  students_sorted.txt  students.txt  vinom.txt
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
vinom.txt: command not found
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
Parvez
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$ sed -i '2i Sahil' vinom.txt
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
Parvez
Sahil
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$ sed -i '2i Ahmad' vinom.txt
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
Parvez
Ahmad
Sahil
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$ sed -i '2a Khan' vinom.txt
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
Parvez
Ahmad
Khan
Sahil
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$ █

```

40. Print only specific lines from a file.

```

vboxuser@ubuntu:~/Desktop/spider$ cd /home/vboxuser/Desktop/spider
vboxuser@ubuntu:~/Desktop/spider$ ls
marks_sorted.txt  marks.txt  studentsl.txt  students_sorted.txt  students.txt  vinom.txt
vboxuser@ubuntu:~/Desktop/spider$ cat vinom.txt
Parvez
Ahmad
Khan
Sahil
Rahul
John
vboxuser@ubuntu:~/Desktop/spider$ sed -n '2p' vinom.txt
Ahmad
vboxuser@ubuntu:~/Desktop/spider$ sed -n '2,3p' vinom.txt
Ahmad
Khan
vboxuser@ubuntu:~/Desktop/spider$

```

41. Copy file from linux to windows machine.

```
USER@DESKTOP-NTBFXNM MINGW64 ~/Downloads
$ ls
Git-2.37.1-64-bit (1).exe*  Mobaxterm_Portable_v22.1/  vc_redist.x64.exe*  desktop.ini  test.pem
Git-2.37.1-64-bit.exe*     Mobaxterm_Portable_v22.1.zip  VSCodeUserSetup-x64-1.134.0.exe*  npp.8.9.7.Installer.x64.exe*  ubuntu-22.04.5-desktop-amd64.iso
IMG-20251226-WA0001.jpg    Slack.msix                 VirtualBox-7.2.16-174877-win.exe*  putty-64bit-0.85-installer.msi

USER@DESKTOP-NTBFXNM MINGW64 ~/Downloads
$ scp -i "test.pem" ec2-user@107.23.59.68:/opt/txt "C:/Users/USER/Downloads/"
scp: open local "C:/Users/USER/Downloads/": No such file or directory

USER@DESKTOP-NTBFXNM MINGW64 ~/Downloads
$ scp -i "test.pem" ec2-user@107.23.59.68:/opt/txt "c:/Users/USER/Downloads/"

USER@DESKTOP-NTBFXNM MINGW64 ~/Downloads
$ op
bash: op: command not found

USER@DESKTOP-NTBFXNM MINGW64 ~/Downloads
$ ls
Git-2.37.1-64-bit (1).exe*  Mobaxterm_Portable_v22.1/  vc_redist.x64.exe*  desktop.ini  test.pem
Git-2.37.1-64-bit.exe*     Mobaxterm_Portable_v22.1.zip  VSCodeUserSetup-x64-1.134.0.exe*  npp.8.9.7.Installer.x64.exe*  txt
IMG-20251226-WA0001.jpg    Slack.msix                 VirtualBox-7.2.16-174877-win.exe*  putty-64bit-0.85-installer.msi  ubuntu-22.04.5-desktop-amd64.iso
```

42. 5 use cases for AWK and 5 use cases for sed.

```
vboxuser@ubuntu:~/Desktop$ cat marks.txt
Parvez 80
Rahul 70
John 90
Sahil 60

vboxuser@ubuntu:~/Desktop$ awk '{sum=sum+$2} END {print sum}'marks.txt
awk: cmd. line:1: {sum=sum+$2} END {print sum}marks.txt
awk: cmd. line:1:                                     ^ syntax error
vboxuser@ubuntu:~/Desktop$ sed -i 's/Rahul70/Rahul 70/' marks.txt
vboxuser@ubuntu:~/Desktop$ cat marks.txt
Parvez 80
Rahul 70
John 90
Sahil 60

vboxuser@ubuntu:~/Desktop$ /usr/bin/gawk '{sum=sum+$2} END {print sum}'marks.txt
gawk: cmd. line:1: {sum=sum+$2} END {print sum}marks.txt
gawk: cmd. line:1:                                     ^ syntax error
vboxuser@ubuntu:~/Desktop$ /usr/bin/gawk '{sum += $2} END {print sum}'marks.txt
gawk: cmd. line:1: {sum += $2} END {print sum}marks.txt
gawk: cmd. line:1:                                     ^ syntax error
vboxuser@ubuntu:~/Desktop$ /usr/bin/gawk '{sum += $2} END {print sum}' marks.txt
300
vboxuser@ubuntu:~/Desktop$ /usr/bin/gawk '{print $1 }' marks.txt
Parvez
Rahul
John
Sahil

vboxuser@ubuntu:~/Desktop$ /usr/bin/gawk '{sum += $2} END {print sum}' marks.txt
300
vboxuser@ubuntu:~/Desktop$ /usr/bin/gawk '{sum += $2} END {print sum}' marks.txt
300
vboxuser@ubuntu:~/Desktop$
```

